

Queue

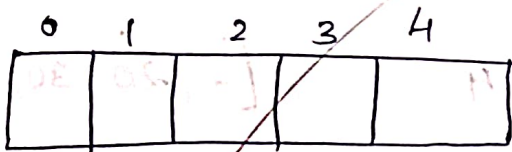
PART - A - Linear Queue

Queue size - 5

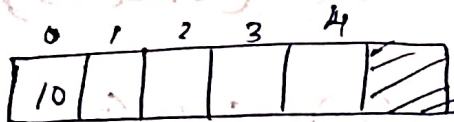
Operations

ENQ(10), ENQ(20), DEQ(),

ENQ(30), ENQ(40), ENQ(50)



Step	operation	Front	Rear	Queue Array
1	ENQ(10)	-1	0	[10, -, -, -, -]



2.

ENQ(20)

-1

1

[10, 20, -, -, -]

0	1	2	3	4
-10	20			

3 $DNQ()$ 0 1 $[-, 20, -, -, -]$

0	1	2	3	4
	20			

4 $ENQ(30)$ 0 2 $[-, 20, 30, -, -]$

0	1	2	3	4
	20	30		

5 $ENQ(40)$ 0 3 $[-, 20, 30, 40, -]$

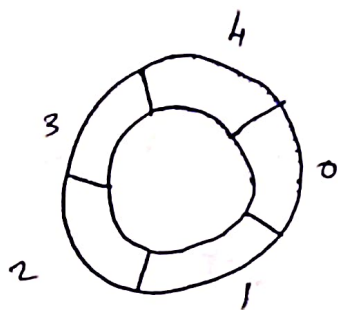
0	1	2	3	4
	20	30	40	

6 $ENQ(50)$ 0 4 $[-, 20, 30, 40, 50]$

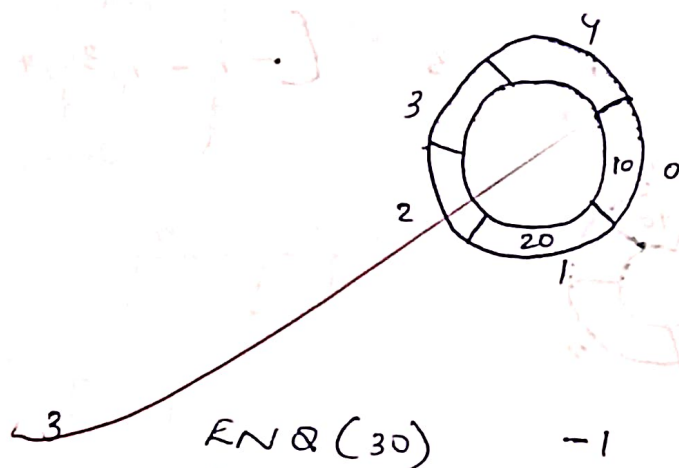
0	1	2	3	4
	20	30	40	50

Part B Circular Queue (size=5)

operations : $ENQ(10)$, $ENQ(20)$, $ENQ(30)$, $ENQ(40)$
 $DEQ()$, $DEQ()$, $ENQ(50)$, $ENQ(60)$



step	operation	Front	Rear	Queue Array
1	ENQ(10)	-1	0	[10, -, -, -, -]
2	ENQ(20)	-1	1	[10, 20, -, -, -]

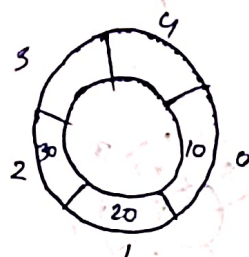


ENQ(30)

-1

2

[10, 20, 30, -, -]

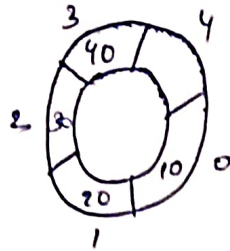


4 ENQ(40)

-1

3

[10, 20, 30, 40, -]



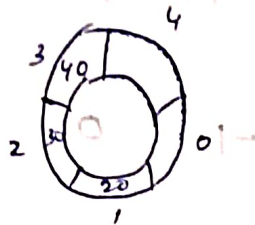
5

ORQ(1)

0

3

$[-, 20, 30, 40,$



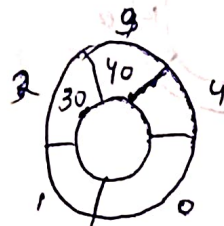
6

ORQ(1)

1

3

$[-, -, 30, 40,$



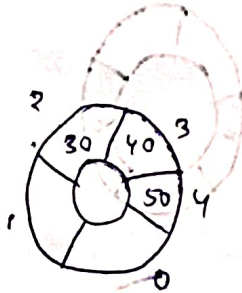
7

KNQ(50)

1

4

$[-, -, 30, 40,$



~~12/12~~

8

KNQ(60)

1

0

$[60, -, 30, 40,$

